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TOPICAL REVIEW

5G/6G Architecture Evolution for XR and Metaverse: Feasibility Study, Security, and Privacy Challenges for Smart Culture Applications

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ABSTRACT This paper investigates the evolution of 5G/6G architectures to support demanding Extended Reality (XR) and Metaverse applications, focusing specifically on the “smart culture” domain. We evaluate the capabilities of the 5G Service-Based Architecture (SBA), including Multi-Access Edge Computing (MEC) and network analytics, through a comprehensive feasibility study comparing stringent XR requirements (bitrate, latency, capacity, power, accuracy) against current 5G performance. Our key contribution is the identification of significant performance gaps where 5G struggles to meet the demands of advanced XR, particularly concerning capacity, scalability, and ultra-low latency. Furthermore, we provide a detailed analysis of critical security and privacy challenges inherent in 5G-enabled XR environments, including virtualization vulnerabilities, API security, and sensitive data protection. While 5G provides core capabilities, significant challenges persist, emphasizing the need for continued research and the evolution toward 6G to effectively support immersive experiences in smart culture and the Metaverse.

INDEX TERMS 5G/6G networks, AR/XR/VR, metaverse, smart culture, education, edge computing, MEC, immersive applications, cybersecurity, privacy.

I. INTRODUCTION

EXtended Reality (XR) technologies, including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), are revolutionizing various sectors, from education to cultural heritage. By merging physical and digital environments, XR provides immersive and interactive experiences that transform the way users engage with information. As the demand for high-quality XR applications grows, network infrastructures must adapt to support the intensive requirements of these technologies, particularly in terms of bandwidth, latency, and reliability.

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5G networks have introduced significant advancements that support the demanding needs of XR, particularly through the implementation of the Service-Based Architecture (SBA) and Multi-Access Edge Computing (MEC). These technologies reduce latency by bringing computational resources closer to users and allow for efficient resource allocation through network slicing and real-time data analytics. While 5G offers a promising framework, there are growing concerns regarding its capacity to fully support large-scale XR applications, especially as user density increases and XR experiences become more technologically complex.

Furthermore, 5G networks expand the potential attack surface, increasing the risk for 5G-enabled XR devices. Vulnerabilities tied to network virtualization and shared

TABLE 1. Scope comparison of papers on XR/Metaverse and their support in 5G/B5G/6G networks.

Ref.	Scope
[1]	The main focus is on 3GPP Rel-18 enhancements and architectural evolution, emphasizing protocol-level solutions such as ECN-based congestion signaling, PDU Set QoS, and DRX-based power saving. It provides a high-level treatment of privacy and security and does not focus on domain-specific applications, such as smart culture.
[2]	Presents a comprehensive 6G-oriented Metaverse architecture with simulation-based insights into remote assistance for autonomous vehicles using 360° streaming. While it offers a well-structured matrix of deployment challenges and solutions, it does not explore vertical-specific contexts, such as smart culture.
[3]	Focuses on a multi-plane (physical, link, virtual) architecture with detailed discussion on enabling technologies like ISAC, SAGSIN, MEC, and blockchain. Prioritizes architectural integration and network-wide coordination across diverse environments, but does not address domain-specific deployments, such as smart culture. Security and privacy are discussed systemically, but not from a user-centric perspective.
[4]	Presents a broad, structured survey of XR-enabling technologies across 3GPP standards and academic literature. Emphasizes taxonomy of solutions, standard KPIs, and architectural models. It is broader in scope than our paper, but does not include empirical feasibility studies or a focused treatment of privacy-preserving mechanisms, such as SSI or smart culture-specific deployments.
[5]	Provides a QoS-centric survey of challenges in Metaverse systems across scalability, security, and integration. Offers future research directions with emphasis on AI/ML, blockchain, and SDN. It does not address specific XR domains or provide empirical evaluation. Identity management and user privacy frameworks are not explicitly explored.
[6]	Focuses on holographic communication as a core enabler for Metaverse experiences, detailing system design, latency and bandwidth requirements, and human-centered evaluation frameworks. Offers a comprehensive architecture and application taxonomy, but does not include empirical benchmarks, domain-specific deployments such as smart culture, or privacy-preserving identity mechanisms.
[7]	Surveys the intersection of AI and 6G across a layered Metaverse architecture, covering learning paradigms, generative content creation, edge AI, and 3D rendering. Provides detailed treatment of enabling technologies and use cases, but does not address domain-specific deployments. Security and governance aspects are discussed mostly in a generalized manner.
[8]	Focuses on cultural heritage as the core application domain, proposing a multi-layer IoT-XR system architecture. Includes real deployments, multisensory interaction (haptics, olfaction), and user evaluations. It does not address 5G/6G architectural evolution or their inherent support for XR.
[9]	Defines the Internet of Cultural Things (IoCT) field and proposes a comprehensive five-layer system architecture. Emphasizes cultural integration via IoT, XR, and edge/cloud computing in tourism, museums, classrooms, and historic districts. While aligned with the cultural domain focus of our paper, it does not frame the discussion in terms of 5G/6G network evolution, protocol-level security, or privacy-preserving identity models.
Our Paper	Focuses on a domain-specific evaluation of XR deployments in smart culture settings, with a focus on empirical feasibility (bitrate, latency, power), threat exposure, and privacy. Highlights security risks tied to MEC, APIs, and Cyber Threat Intelligence sharing (CTI), and explores user-centric design through SSI wallets and Digital Twins. Offers a complementary perspective to standards-focused works by grounding architectural evolution in application-driven requirements.

Multi-Access Edge Computing (MEC) resources pose new threats. XR devices, equipped with various smart sensors, collect vast amounts of personal data, such as user movements, physiological details, and interactions in virtual spaces, making them appealing targets for data leaks, unauthorized profiling, and data interception. Moreover, XR applications face specific risks from network-level attacks, such as those exploiting Radio Resource Control (RRC) or GTP protocols, which can lead to session hijacking or denial of service, especially in real-time environments. With the growing adoption of XR technologies, these security and privacy risks emphasize the need for strong countermeasures and comprehensive data protection protocols.

A. RELATED WORK

Recent research has extensively explored the convergence of 5G/6G networks with Extended Reality (XR) and Metaverse technologies, with particular attention to enabling seamless, low-latency, and immersive experiences. We surveyed key studies between 2023–2025, shown in Table 1, to provide a

comprehensive view of the state-of-the-art in XR/Metaverse integration with mobile networks. These papers were selected based on their thematic relevance, technical diversity, focus on real-world deployment scenarios, and inclusion of cultural, security, or user-centric perspectives.

Several papers emphasize advances in 5G/6G network architecture. In [1], the authors examine 3GPP Rel-18 enhancements that support XR workloads, such as PDU Set-based QoS and DRX optimization, offering a standards-aligned view of architectural evolution. In [3], the authors provide a roadmap for integrating Metaverse services across physical, link, and virtual planes using enabling technologies, including ISAC and SAGSIN. The authors in [4] offer a structured taxonomy of XR service enablers across 3GPP releases and academic literature, covering solutions across application, edge, and core network layers.

Other works focus more on architectural or system-level design. The work in [2] proposes a layered 6G Metaverse framework with simulation-based insights into remote vehicle assistance using real-time VR. The authors

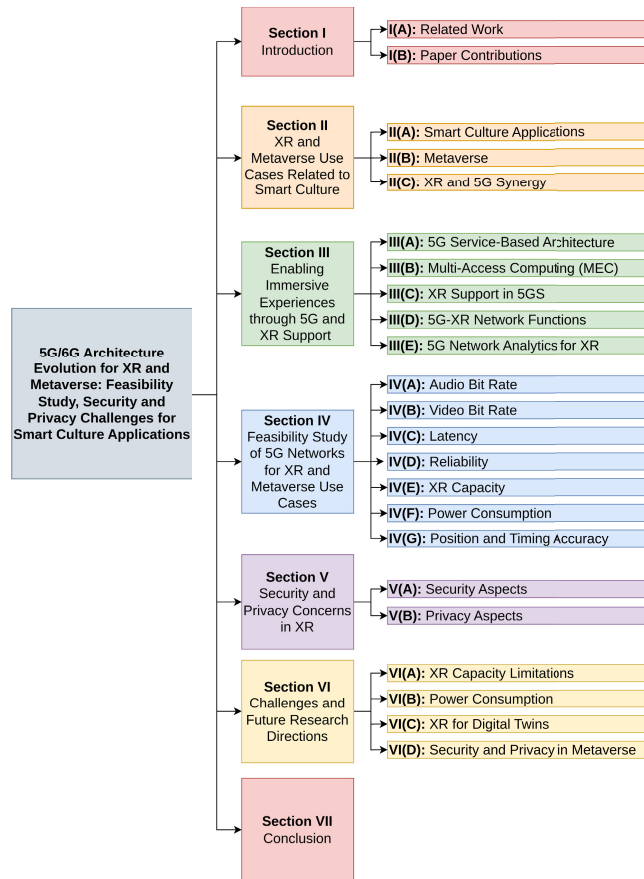


FIGURE 1. Paper structure.

in [6] highlight holographic communication as a foundational technology for immersive Metaverse applications, identifying key bandwidth and latency demands. The work in [7] surveys how AI techniques, including generative models and edge inference, contribute to immersive Metaverse services, framed within a layered architecture derived from game design and social interaction models.

Other contributions prioritize specific technical dimensions [5]. The authors analyze the role of 5G and 6G in meeting Metaverse QoS requirements, such as ultra-low latency, high throughput, and scalability. Their work identifies research gaps in system integration, authentication, and resource orchestration. While they emphasize infrastructure requirements, they do not target domain-specific deployments.

Several works bring cultural and user-centric applications into focus. The authors in [8] review IoT-XR integration in cultural heritage through 3D reconstruction, intelligent tours, and multisensory interaction. In [9], the authors introduce the concept of the Internet of Cultural Things (IoCT), discussing edge-cloud architectures for museums, historic districts, and educational experiences. Although these papers align with the cultural focus of our work, they do not consider 5G/6G network evolution or protocol-level privacy mechanisms.

These papers provide valuable insights into enabling technologies, architectural frameworks, and emerging challenges

in supporting Metaverse and XR services over future mobile networks. However, they either focus on generalized system architecture, theoretical frameworks, or application-specific implementations without connecting network evolution to security and privacy risks within domain-specific XR deployments.

B. PAPER CONTRIBUTIONS

This paper contributes a complementary perspective to the existing literature by focusing on the current capabilities of 5G networks to support XR applications in smart culture scenarios. Unlike works that examine either abstract architectural models or anticipated 6G functionalities, our paper assesses whether today's 5G infrastructure—particularly its Service-Based Architecture (SBA) and Multi-Access Edge Computing (MEC)—can meet the demands of immersive, interactive cultural experiences.

By comparing key performance indicators (bitrate, latency, power, and reliability) with the requirements of typical XR workloads, we present a detailed feasibility study grounded in real-world constraints. In addition, our work uniquely explores security and privacy risks in these deployments, including threats to virtualized infrastructure, MEC exposure, and identity leakage, and proposes mitigation strategies such as Cyber Threat Intelligence (CTI) sharing and Self-Sovereign Identity (SSI) wallets.

Our main contributions are summarized as follows:

- A comprehensive exploration of 5G's role in enabling XR applications, particularly for smart culture use cases. It examines how 5G SBA and MEC support low-latency, real-time interactivity while emphasizing the role of network analytics in optimizing resource allocation.
- A detailed assessment of performance metrics—including bitrate, latency, reliability, and power consumption—against XR and Metaverse application requirements, grounded in both literature and practical observations.
- A focused analysis of security, privacy, and ethical concerns associated with immersive XR technologies, proposing identity-centric safeguards and network-integrated trust mechanisms.

Fig. 1 illustrates the structure of this paper. Section II introduces XR and Metaverse use cases. Section III details the 5G SBA architecture. Section IV assesses the feasibility of 5G for XR deployments. Section V discusses security and privacy. Section VI outlines future challenges. Section VII concludes the paper.

II. XR AND METAVERSE USE CASES RELATED TO SMART CULTURE

The integration of XR technologies with smart culture is opening up new possibilities for enhancing user experiences in entertainment, education, and cultural heritage. Some of the most promising use cases include hybrid experiences, remote exhibitions, virtual events, and collaborative art

initiatives. Key requirements for relevant network and connectivity technologies are related to the scalability of heterogeneous devices that should be interconnected to support the required data flows and the computation capabilities that should support the smart culture applications. Centralized cloud computing topologies may not address requirements for real-time processing and sense-decide-act loops, involved in immersive applications. In contrast, distributed edge-based architectures could provide the required solutions for localized data processing and reduced latency, at the expense of more complex distributed cloud edge continuum management frameworks [8].

A. SMART CULTURE APPLICATIONS

Mixed reality enables the fusion of real and digital environments, creating hybrid experiences. Museums are using AR apps linked to QR codes to immerse visitors in exhibits, while entertainment companies stage holographic projections of artists, making live concerts a reality for remote spectators. These technologies also improve accessibility, offering features such as real-time sign language and translation options [10]. Remote exhibitions and museums leverage AR to bring art into users' homes. Visitors can visualize artwork on their walls using mobile apps, enabling a seamless connection between art and e-commerce [11].

Virtual events are transforming how audiences engage with performances and education. Artists, such as musicians and comedians, use XR to reach global audiences. These platforms also offer educational benefits, enabling students to explore virtual environments as complementary learning tools, regardless of physical limitations [12]. Creative collaboration has been enhanced with XR technologies. Content creators across the globe can now produce music videos and immersive experiences using VR and MR, signaling a future where virtual collaboration becomes standard in the arts.

With real-time virtualization, museums offer dynamic exhibits where users receive personalized information as they move through physical spaces. The growing demand for resource-intensive applications, such as large-scale metropolitan art installations, requires high-speed data rates and low-latency capabilities [13]. The Interactive, Responsive Museum Experience framework, an open-source IoT-enabled system, integrates XR with VR, AR, haptic interfaces, and 3D imaging to engage visitors on a deeper level. This allows a broader audience, including individuals with disabilities, to interact with exhibits through multisensory experiences [14].

In addition, China's Hunan Museum [15], with full 5G coverage, showcases innovations such as virtual tours and artifact interactions, 4K/8K live streaming of historical artifacts and holographic displays for historical reconstructions.

Furthermore, the CHRONOS Application reconstructs virtual digital representations of notable landmarks in Ancient Greece such as the Parthenon, the Temple of Artemis Brauronia, the Chalkotheke (which no longer exists today),

and the Odeon of Herodes Atticus, all viewed from the perspective of the Acropolis Museum [16].

Additionally, relevant applications have been proposed based on AR visual calibration combined with barometric pressure data for indoor localization and navigation [17], or real-time 3D scanning and rendering [18] that may also be assisted by IoT networks [19] for identifying the visitors' movement patterns and for providing targeted multimedia or AR/VR content related to the monitored artifacts and also for collecting important information about the area environment. This integration of intelligent devices, sensors, cameras and AR/VR enablers for smart culture applications has led to the definition of the Internet of Cultural Things (IoCT) [9]. Such applications may also combine user preferences and usage profiles with environmental conditions for generating virtual tours and personalized paths within the visited place, enriching thus a visitor's cultural experience and providing a whole new level of engagement and flexibility [20]. It is therefore evident that technologies such as AR, VR, 5G and the Internet of (Cultural) Things can uplift cultural heritage and promote its sustainability as well as accessibility to the public [21].

B. METAVERSE

The Metaverse is considered the next evolution in digital interactions and merges physical and virtual worlds into a persistent, interconnected digital platform. Although still in development, multiple metaverses already exist today, enabling virtual experiences across sectors such as health-care, education, smart cities, and entertainment [22].

The Metaverse has the potential to enhance cultural heritage preservation through:

- Immersive experiences: Recreating historical sites and events in lifelike virtual environments.
- Unconditional participation: Providing access to cultural heritage content from anywhere, at any time.
- User-Generated Content: Facilitating the co-creation of cultural experiences and exhibits by participants.
- User-Generated Content: Facilitating the co-creation of cultural experiences and exhibits by participants.
- Autonomous Economy: Allowing creators to develop and trade digital assets and experiences within a virtual world [23].

The IoT ecosystem plays a significant role in the Metaverse by enabling real-time, context-aware XR applications. Through IoT data and Tactile Internet capabilities, the Metaverse can react to users' physical gestures and environmental changes, creating a seamless interaction between the digital and physical worlds. 5G technology, with its high-speed, low-latency architecture, provides the foundation for these immersive experiences [24], [25].

5G's role is particularly important in cultural heritage, where it enables large-scale, real-time interactions within the Metaverse. Whether recreating ancient landmarks or hosting

virtual exhibitions, 5G empowers the dynamic exchange of rich content across multiple users [26].

Cultural heritage involves diverse stakeholders with various interests, which can present challenges for digitalizing this sector [27]. The development of a Metaverse for cultural heritage must incorporate intricate content, representations, and expressions tied to unique cultural items. Shared digital platforms should cater to all stakeholders' needs, despite the diversity in cultural heritage expressions.

Previous research [28] presents an early example of using a digital interface to create a 3D virtual excavation monograph, linking physical and digital spaces. Other studies, such as [29], emphasize the importance of linking virtual environments to remote datasets for comparative analysis. Both examples illustrate the importance of integrating physical and virtual worlds in cultural heritage through XR and Metaverse technologies.

In terms of cultural heritage representation, various forms are available, ranging from sequential symbols in folk music to 2D visual art and 3D sculptures. Digital representations of cultural heritage could enhance storytelling and time-varying narratives by incorporating these formats into virtual environments [26].

C. XR AND 5G SYNERGY

The aforementioned use-cases highly benefit from the deployment of 5G technology which is revolutionizing AR, MR, and VR by delivering high-speed data transmission, low latency, and multi-device connectivity. These features are essential for enabling high-resolution content, real-time interactivity across XR platforms [30] and collaborative events. In detail, 5G synergy offers:

- **High-Resolution Experiences:** 5G supports the quick and efficient delivery of complex visual content required by AR/MR/VR.
- **Real-Time Interactions:** Low latency in 5G ensures smooth and seamless interactive experiences.
- **Social Experiences:** 5G enables multi-user interactions within virtual environments, allowing users to engage in real-time, synchronized activities.

The 5G-enabled XR use cases extend to cultural heritage and education, providing new ways to share, preserve, and interact with historical and educational content. Key applications include [30]:

- **Offline Sharing of 3D Objects:** Users can exchange 3D models and mixed reality scenes for collaborative projects.
- **Real-Time XR Sharing:** Devices can capture and stream XR experiences in real time to enhance educational and cultural interactions.
- **XR Conferences:** These platforms blend physical and virtual spaces, allowing in-person and remote participants to share a unified experience.
- **XR Multimedia Streaming:** Users can stream 2D or 3D XR content in real-time or on-demand, with the

ability to control playback and interact within the virtual environment.

III. ENABLING IMMERSIVE EXPERIENCES THROUGH 5G AND XR SUPPORT

This section provides an overview of the 5G Service-Based Architecture (SBA), detailing its design choices and capabilities. It highlights ongoing efforts by 3GPP to integrate Extended Reality (XR) support into the 5G SBA. Additionally, the section examines key characteristics of 5G networks that enable the XR use cases discussed in Section II, and assesses the feasibility of these use cases by analyzing relevant XR performance metrics.

A. 5G SERVICE-BASED ARCHITECTURE (SBA)

The Service-based Architecture for 5G Core introduced new features to meet the diverse demands of modern telecommunication systems, which legacy networks cannot adequately address. Such demands include stringent service requirements for specialized industries, *e.g.*, Automotive Industry, XR applications, Smart Cities and IoT. 3GPP designed the 5G SBA to provide a modular design that enables rapid service deployment based on cloud-native principles. The 5G SBA features are summarized as follows:

- **Network Functions (NFs):** In the 5G SBA, the traditional entities of legacy cellular networks are decomposed into distinct NFs, each with specific features, protocols and responsibilities, such as user authentication and session management.
- **Stateless Design:** NFs are designed to be stateless. Session states are managed by designated data repositories, such as the Unified Data Management (UDM).
- **HTTP/2 Protocol:** SBA leverages HTTP/2 for communication among NFs, reducing latency and improving network efficiency.
- **Network Slicing:** SBA supports network slicing, allowing the creation of several virtual networks over a shared physical infrastructure. Each slice is configured to address specific service requirements, such as eMBB, uRLLC. Network slicing was introduced as a native feature of the 5G SBA.
- **API-driven:** The 5G SBA provides the standardized exposure of network capabilities to third-party developers, supporting the development of new and innovative applications that leverage 5G NFs.
- **Multi-Access Edge computing (MEC):** The 5G SBA aligns with MEC principles and facilitates the deployment of network resources closer to the user allowing the deployment of NFs at the edge.

SBA architecture includes a set of NFs that interact with each other over standardized interfaces. The NFs are interconnected through service-based interfaces (SBIs) that use HTTP/2 as the communication protocol, and each NF acts as a consumer or producer of services. 3GPP defines

the interactions among NFs to ensure interoperability and consistency across different vendors.

B. MULTI-ACCESS EDGE COMPUTING (MEC)

Multi-access Edge Computing (MEC) is a network architecture concept that brings computational resources, data storage, and applications closer to the user at the edge of the network. In the context of 3GPP's 5G New Radio (5G-NR), MEC is implemented to support ultra-reliable low-latency communication (URLLC), enhanced mobile broadband (eMBB), and massive machine-type communications (mMTC), which are the three primary service types defined by 3GPP for 5G.

MEC in 5G NR is realized through the deployment of edge servers that are integrated with the 5G core network, as depicted in Fig. 2. 3GPP TR 23.758 Rel. 17 [31] and TS 23.558 Rel. 17 [32] introduce application layer interfaces for MEC, facilitating the discovery of edge servers, exposing capabilities to these servers, and managing the lifecycle of edge applications. These servers host functions and applications that traditionally reside in centralized data centers. By decentralizing these functions, MEC reduces the end-to-end data distance, reducing latency and improving service delivery. 3GPP has standardized the interfaces and protocols for MEC integration within the 5G system, as depicted in Figure 1. This includes the definition of service-based architectures (SBA) for the 5G core, which allows network functions to be modular and interoperable. MEC leverages these SBAs to interact with core network functions such as the User Plane Function (UPF) for traffic routing and the Session Management Function (SMF) for session control.

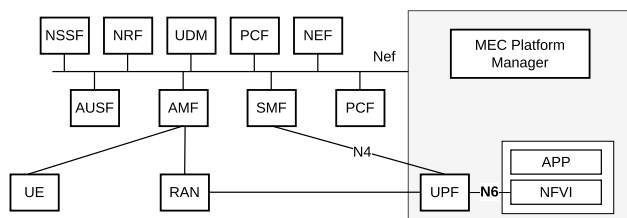


FIGURE 2. MEC integration within the 5G SBA [32].

Furthermore, 3GPP has worked on defining the requirements and frameworks for network exposure functions (NEF), which enable MEC applications to access network capabilities and information. This allows MEC applications to be context-aware and to adapt to real-time network conditions and user needs.

In MEC, the User Plane Function (UPF) and Data Networks (DN), including Local Area Data Networks (LADN), are located near the base station, which could be integrated with the station itself or a nearby router. This reduced proximity can enable Extended Reality (XR) and cloud gaming (CG) services that require immediate data processing for real-time analytics. For example, a museum could leverage Augmented Reality (AR) to overlay digital

information on exhibits, with the edge computing server tracking visitors' locations to deliver tailored content.

In addition, edge computing supports content caching for rapid streaming of high-demand XR content. Such content may include interactive features, audio tracks, commentary, popular camera angles, highlights, and events—all of which can be cached at the edge based on user demand and engagement patterns.

C. XR SUPPORT IN 5GS

3GPP began formalizing XR support through NR in 2016 when the Service and System Aspect (SA) working group on "Services" introduced the 5G service requirements for XR applications that demand high data rates and low latency. This effort continued in 2018 with SA4 ("Multimedia Codecs, Systems and Services") detailing the traffic characteristics for XR in TR 26.925 Rel. 18 [33] and surveying XR applications. SA ("System Architecture and Services") established new 5G Quality of Service (QoS) identifiers (5QIs) to cater to interactive services, including XR, in TS 23.501 Rel. 18 [34]. Release 17 introduced a unified set of traffic models and an evaluation framework for XR, while preliminary studies on potential XR solutions began in the early phase of Release 18.

3GPP's Release 18 introduces the XRM Work Item to enhance 5G capabilities for XR and media services. This includes improvements for multi-modality transmission, network information exposure for media service adaptation, and Packet Data Unit (PDU) Set-based QoS handling for precise media unit delivery. It also features uplink-downlink coordination to meet real-time latency requirements, Packet Delay Variation monitoring for consistent network performance, and power-saving features for XR devices. Policy control is updated to manage various data types for synchronized delivery. Network information exposure uses Explicit Congestion Notification (ECN) marking and APIs for media adjustments. The PDU Set concept is designed for media-specific QoS control, while transmission coordination and PDV monitoring ensure service quality. Energy-saving strategies are developed for the intermittent nature of media services, supporting battery-limited XR devices.

D. 5G-XR NETWORK FUNCTIONS

In 3GPP TR 26.928 Rel. 16 [30], the WG introduced a potential 5G architecture that integrates recommended 5G-XR specific Network Functions, as illustrated in Fig. 3. In this report, there are several recommendations for enhancing the existing 5GS as defined in TS 23.501 [34], by introducing XR-specific NFs. The architecture serves as a foundation, with potential extensions to accommodate XR applications, potentially leveraging network slicing, edge computing, and enhanced 5G quality of service to improve XR application performance.

A 5G-XR application provider offers services utilizing 5G system functionalities through a 5G-XR aware application

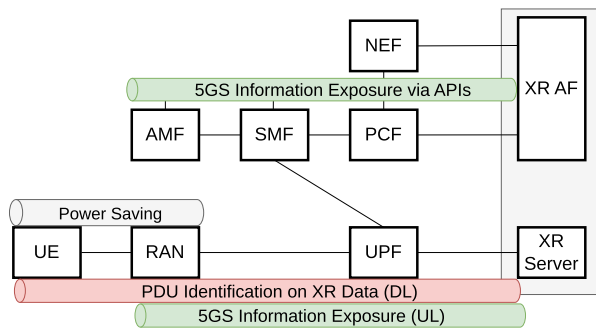


FIGURE 3. XR support over 5GS based on 3GPP [32], [35].

on user equipment (UE), interacting with a 5G-XR client and network functions via specified interfaces and APIs. The architecture, as depicted in 3GPP TS 23.501 [34], includes three primary functions: the 5G-XR Application Function (AF) for XR services, the 5G-XR Application Server (AS), and the 5G-XR Client within the UE (Fig. 3). These functions, considered part of the Data Network (DN), communicate with the UE through N6, N3, and Uu interfaces, with sidelink PC5 as a possible communication alternative. Trusted DN functions are deemed secure by the operator's network, allowing direct communication with all 5G Core functions. In contrast, external DN functions must communicate through the Network Exposure Function (NEF) using N33.

E. 5G NETWORK ANALYTICS FOR XR

The integration of Artificial Intelligence (AI) and Machine Learning (ML) into 5G-Advanced is progressing rapidly and could be used to optimize networks for XR services. 3GPP, in Release 18, is examining AI/ML improvements for the lower layers of the air interface. At the same time, techniques for the higher layers in the RAN and SA have been progressively developed since Release 16, and now include AI/ML-specific operations such as data collection and model training. AI/ML is a key enabler for analyzing XR traffic patterns, facilitating smarter application-aware scheduling, and developing intelligent tools for Discontinuous Reception (DRX) to increase capacity and improve energy savings.

The Network Data Analytics Function (NWDAF) was introduced as part of the 3rd Generation Partnership Project (3GPP) Release 15 specifications and represents a shift towards data-driven cellular networks. The NWDAF is designed to aggregate and analyze data from various sources within the 5G network. These data include performance metrics, user behavior, and network conditions. The insights derived from this analysis are used to optimize network performance, manage resources more efficiently, and improve user experiences. The NWDAF is integrated into the Service-Based Architecture (SBA) of 5G, interacting with other network functions via standardized interfaces, as depicted in Fig. 4. Consequently, data are collected from various network elements in order to provide analytics

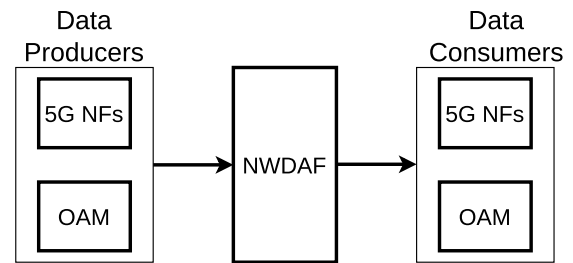


FIGURE 4. NWDAF in 5G [32].

services to other functions like the Network Slice Selection Function (NSSF) or the Policy Control Function (PCF).

As a result, the NWDAF can predict network demands, detect anomalies, and provide recommendations for network configuration changes. NWDAF supports a range of use cases, from abnormal user behavior to quality-of-service optimization. As a result, it is considered a significant enabler for advanced 5G use-cases with diverse service requirements, including XR Use Cases. XR Use Cases require high throughput and low latency to deliver the desired performance and user experience. The reactive nature of legacy cellular networks cannot guarantee optimal resource allocation of the network to address the dynamic and ever-changing traffic demands of such immersive use cases.

NWDAF may employ AI/ML algorithms to analyze XR traffic patterns, enabling it to predict network load and adapt to XR service demands in real-time. It can also assist in network load balancing by identifying high XR usage areas, such as museums, to optimally allocate network resources, manage bandwidth, and network slicing, ensuring XR applications are adequately supported without overloading the network. In QoS management, NWDAF can dynamically adjust parameters for XR traffic, prioritizing data packets for efficient routing with minimal delay and packet loss. Moreover, traffic can be directed to the nearest edge server based on network conditions and user locations, reducing latency and enhancing the overall user experience.

IV. FEASIBILITY STUDY OF 5G NETWORKS FOR XR AND METAVERSE USE CASES

This section evaluates the readiness and capability level of 5G networks to support the demanding requirements of XR and Metaverse use cases. Table 3 outlines key performance metrics necessary for XR and Metaverse applications, such as audio and video bit rates, latency, reliability, power consumption, capacity, and positioning accuracy. Table 2 summarizes the assumptions extracted from the papers in Table 3. Consequently, performance metrics are compared against the current capabilities of 5G networks extracted from the literature to highlight both strengths and limitations in supporting smart culture applications and use cases.

A. AUDIO BIT RATE

According to [1], the current audio standard in XR applications typically operates at around 36 Mbps, but the projected

TABLE 2. Evaluation assumptions of papers in Table 3.

Ref.	Assumptions
[36]	- XR and eMBB coexistence in 5G-A networks using two 3GPP-aligned deployment scenarios. Indoor Hotspot (InH): Area: 120×50 m with 12 ceiling-mounted low-power gNBs (20 m ISD). UEs uniformly distributed. 3 km/h mobility. XR density: 1 UE per 167 m^2. Dense Urban (DU): 3-sector outdoor gNBs in a hexagonal grid (200 m ISD). 80% indoor UEs distributed across 6-floor buildings; 20% outdoor at 1.5 m height. 12° antenna downtilt; XR density: 1 UE per 2891 m^2. - XR traffic: 60 fps, 45 Mbps stream (750 kbits/frame). Jitter modeled as truncated Gaussian: $\mu_j = 0 \text{ ms}$, $\sigma_j = 2 \text{ ms}$. eMBB traffic: full-buffer model
[37]	3GPP-aligned macro topology: 7 sites with 21 cells, 500 m inter-site distance (ISD). - Antenna configuration: 8 Tx / 2 Rx antennas. Carrier frequency: 2.4 GHz with 40 MHz bandwidth. - Channel model: 3GPP-3D fading. Traffic model: FTP3 (1 MB packets, Poisson arrival). Scheduler: proportional fair. HARQ with Chase Combining. Uplink power control: 3GPP-compliant.
[1]	- Proposes an integrated 5G-Advanced architecture for XR with end-to-end application–network coordination aligned with 3GPP Releases 17 and 18. - XR service requirements including: Lossless audio transmission: 1.6 Gb/s. Ultra-high-resolution video: up to 1 Tb/s (post-compression). Motion-to-sound delay: < 20 ms. FoV refresh latency target: < 14 ms (1 ms pose sensing + 5 ms rendering).
[38]	- XR loopback mechanism to dynamically adapt XR traffic based on 5G network conditions using an open-source 5G system-level simulator. - Simulation setup includes multiple XR UEs and eMBB traffic with FTP3 traffic models, 1 gNB (Central Frequency: 4GHz, Bandwidth: 80MHz, Channel Model: 3GPP TR 38.901 UMa)
[39]	- Focuses on split XR over 5G using edge rendering and HMD feedback loops. - XR downlink traffic modeled at 30–45 Mbps, 60 fps. XR frame size follows truncated Gaussian distribution: Mean: μ, Standard deviation: $\sigma = 10.5\% \mu$, bounds: 0.5μ to 1.5μ. Jitter modeled as truncated Gaussian: $\sigma = 2 \text{ ms}$, range: $[-4 \text{ ms}, 4 \text{ ms}]$. - Latency requirements: Motion-to-render-to-photon (m2r2p): < 100 ms (target: 50–100 ms). Round Trip Time (RTT): < 20 ms. Simulation setup based on 3GPP TR 38.838: Dense Urban (FR1) and Indoor Hotspot (FR2). Power constraints: < 3 W for XR devices; < 0.5 W for 5G modems.
[40]	- Uses a 3GPP-calibrated Nokia system-level simulator. Full protocol stack simulated: PHY to PDCP, including MIMO, HARQ, scheduling, and inter/intra-cell interference. - Deployment scenarios: Indoor Hotspot (InH): 12 indoor cells, 50×120 m area. Dense Urban (DU): 21 macro cells, 530×460 m area. - UEs randomly placed with speed: 3 km/h. Scheduler: proportional fair. - Carrier frequencies and bandwidth: FR1: 4 GHz; FR2: 30 GHz; 100 MHz bandwidth. XR traffic model: DL: 60 fps video. UL: 250 fps pose updates, 100 fps audio/data, optional AR video stream.
[41]	- Focuses on architectural enablers and technical mechanisms for providing time synchronization as a service in 5G-Advanced and future 6G networks. - No explicit simulation is conducted, but the paper outlines timing synchronization methods including Access Stratum Time Information (ASTI), generalized Precision Time Protocol (gPTP), and PTP, with synchronization accuracy targets ranging from 5–10 μs for basic services to sub-μs for direct air interface delivery. - Assumptions include centralized coordination via the Time Sensitive Communication and Time Synchronization Function (TSCTSF) and support for temporal and spatial validity conditions in timing service APIs.
[42]	- Scenario: $10 \times 10 \times 2.5$ m indoor factory-like environment based on 3GPP evaluation assumptions. - XR device equipped with 5 antennas in a fixed constellation, height: 1 m. - TRP deployment: 3 TRPs at fixed coordinates operating at 3.5 GHz (FR1) and 28 GHz (FR2). - Ranging error standard deviation: $\sigma_r = 50 \text{ mm}$ (FR1), $\sigma_r = 5 \text{ mm}$ (FR2). - Duration: 30 s, sampled every 10 ms (1 NR frame). Human-like velocity up to 4 km/h; orientation change up to $90^\circ/\text{s}$. 100 Monte Carlo test tracks per scenario. Integer ambiguity and UE clock drift explicitly modeled.

requirement for lossless audio is 1.6 Gbps, a much higher threshold. The authors note that while 5G can handle standard audio rates, it cannot address the demands of lossless audio in XR environments, particularly as applications scale in complexity and user interactions. This gap indicates a substantial challenge in supporting the immersive, high-fidelity audio experiences that XR platforms aim to deliver.

Furthermore, the paper highlights that while 5G networks can achieve high-quality audio at these standard rates, they are not yet capable of consistently delivering the 1.6 Gbps required for future XR audio demands, especially in dense, high-traffic environments.

In [37], experiments on user satisfaction with XR services under different data rate conditions showed how increasing bit-rates impact user experience. For XR services such as

AR, VR, and cloud gaming (CG), at a data rate of 30 Mbps, the satisfaction rate for XR users per cell is approximately 95%, with the system able to support about 4.2 users per cell. However, when the data rate is increased to 45 Mbps, the number of satisfied users drops drastically to just 1.6 users per cell, indicating a significant decline in satisfaction as bit-rates increase. These data support the observation that, while 5G can handle high-quality audio, user satisfaction decreases as data rates increase, since the network struggles to maintain the necessary quality of service for a larger number of users at higher bitrates.

Although 5G provides substantial improvements in bandwidth and latency compared to previous cellular networks, supporting the future demands of lossless audio transmission remains a challenge.

TABLE 3. Requirements for metaverse use cases and evaluation of support over 5G.

Key XR Metric	Requirement	5G Support
Audio Bit Rate	Lossless audio bit rate projected to be 1.6 GB/s. Current audio standard is 36 Mb/s. [1]	5G supports high-quality audio, but cannot support the 1.6Gb/s bitrate required for lossless audio in XR [1]. - Experiments in [37] indicate significant drops in user satisfaction as data rates increase.
Video Bit Rate	Full-view high-resolution video transmission is estimated at 1 Tb/s. The current video bit-rate for 60 PPD resolution is 1 Gb/s. [1]	5G supports high-resolution video, but in dense environments, data rates above 30 Mbps significantly reduce user satisfaction [37]. - Larger PRB utilization ($\geq 50\%$) for eMBB traffic leads to delays exceeding the 10 ms PDB, impacting XR user satisfaction [36]. - The AIPAT algorithm achieves up to 50 Mbps for VR and 40 Mbps for cloud gaming with low packet loss and maintains packet delay under 8 ms [38]. - XR loopback mechanism improves throughput by 25-50% in dynamic network conditions [38].
Latency	XR requires a motion-to-sound delay of less than 20 ms. Motion-to-render-to-photon latency should be under 20 ms to avoid motion sickness. [1]	5G URLLC meets latency needs (<10 ms), but as user density increases, buffering delays affect user satisfaction [37]. - When user density increases, motion-to-photon latency for XR applications can reach up to 50-100 ms, impacting the XR experience, especially in densely populated environments [39].
Reliability	For XR applications, high reliability refers to ensuring that a high percentage (e.g., 99%) of video frames are delivered within the target latency, maintaining user satisfaction. [39].	5G URLLC provides high reliability, but performance drops significantly in high-density scenarios [37]. - In dense urban environments, 5G supports 13 UEs per cell with 30 Mbps XR traffic, but satisfaction rates drop as more users are added. - Interference mitigation schemes, such as ICIC and restricting eMBB PRB utilization, improve XR reliability in mixed traffic scenarios [36].
XR Capacity	The maximum number of UEs per cell with at least Y% of the UEs being satisfied [40].	- In frequency range 1 (FR1), 5G supports 2-5 UEs per cell with 45 Mbps XR traffic and 3-8 UEs per cell with 30 Mbps traffic using traffic staggering [39]. - In FR2 (400 MHz), capacity improves from 20 to 28 UEs for 45 Mbps and from 26 to 35 UEs for 30 Mbps traffic [39]. - Simulations show 5G can support 6 XR devices per cell with 30 Mbps traffic in dense urban environments (FR1), but capacity drops non-linearly with increased data rates or strict latency requirements [40]. - Increasing PDB for cloud gaming services to 15 ms improves capacity by 1.3x compared to AR/VR services with 10 ms PDB [40]. - eMBB traffic reduces XR capacity by up to 80%, but with radio resource management (RRM), XR capacity can be maintained [36]. - In dense urban scenarios, XR capacity drops to one satisfied user per cell when eMBB traffic is present [36].
Power Consumption	Low power consumption devices ($<3W$) required for HMDs and AR glasses. [39]	5G modem power consumption ($0.5W$) is supported, but reducing power with low latency and dense traffic remains a challenge [39]. 5G-A introduces DRX cycle optimization based on media stream periodicity (e.g., 60 fps video) and end-of-burst detection to further save power without impacting traffic delivery [1].
Positioning & Timing Accuracy	Precise location tracking for AR/VR experiences, indoors and outdoors. Accurate timing synchronization among multiple XR devices for Metaverse use cases.	5G-A achieves microsecond-level accuracy ($1 - 10\mu s$), through synchronization Over-The-Air (OTA), Precision Time Protocol (PTP) and Time Sensitive Networking (TSN) [41]. - Six degrees-of-freedom (6DoF) tracking of XR devices (3D location and 3D orientation) for real-time tracking of XR devices. The method on [42] uses 5G UL carrier phase measurements to achieve RMSE can provide positioning accuracy within 1 cm and orientation accuracy below 1 degree. XR positioning accuracy is feasible down to sub-centimeter level (0.1 mm to 5 mm) at higher frequency bands (such as 28 GHz) [42].

In XR services, where synchronization between video and audio is critical, the audio transmission must maintain a delay of less than 20 ms to prevent noticeable desynchronization between motion and sound. A challenge like this becomes more pronounced when considering the need for precise timing in 3D space, where any jitter or latency could degrade the immersive experience. Thus, the demands of lossless audio are often underestimated but must be re-evaluated alongside other high-data-rate media,

such as 4K video, to ensure an optimal user experience in XR.

B. VIDEO BIT RATE

Video transmission for high-resolution XR experiences places significant demands on network bandwidth. According to [1], achieving full-view, high-resolution video (with a resolution of 72K by 36K) across a 360-degree horizontal and 180-degree vertical field of view requires an estimated

bit rate of 1 Tb/s. Even with advancements in video compression, such as the H.266/VVC standard, which can reduce the required bit rate by a factor of 500, this still translates to a current requirement of approximately 1 Gb/s for XR applications with 60 PPD resolution [1]. This makes supporting ultra-high-resolution XR video transmission one of the most demanding tasks for future network technologies, including 6G.

However, current 5G systems are capable of supporting lower-resolution XR video transmissions. For example, the authors in [37] show that, in dense environments, maintaining data rates above 30 Mbps for high-resolution video significantly reduces user satisfaction. This indicates a limitation in 5G networks when handling high-bandwidth XR video traffic in environments with many users. To address these challenges, approaches like resource optimization and scheduling enhancements are needed to improve user experience for XR applications over 5G and beyond networks.

In addition, the authors in [38] report that their proposed AIPAT algorithm achieves up to 50 Mbps for virtual reality (VR) and 40 Mbps for cloud gaming, with packet loss kept to a minimum and packet delay consistently maintained under 8 ms. This helps ensure smooth video streaming under fluctuating network conditions by dynamically adapting traffic.

Despite these advancements, however, delivering ultra-high-resolution XR content will continue to be a challenge, and further research into network optimization and video compression is needed to meet future demands.

C. LATENCY

XR experiences demand extremely low latency, especially to avoid motion sickness. Motion-to-sound delays should remain below 20 ms, and the critical motion-to-render-to-photon latency (the time from a user's movement until the corresponding frame is rendered and displayed) must also remain under 20 ms for XR systems to offer a seamless experience [1].

5G URLLC (Ultra-Reliable Low Latency Communication) can meet these requirements by offering sub-10 ms latencies in optimal conditions, enabling real-time interactivity. However, in dense environments -like urban settings or museums- where many users may share the network, buffering delays and congestion can increase latency significantly, sometimes reaching up to 50–100 ms, which may disrupt user experiences and cause discomfort. Therefore, as user density increases, latency grows, especially in XR applications. For example, according to [39], motion-to-photon latency can jump to 50–100 ms in high-density environments, negatively impacting XR user experiences. Such delays are problematic in scenarios like interactive exhibits in museums, where user interactions must be real-time to be effective.

Although 5G provides sufficient performance in some cases, as discussed in [37], increasing traffic and PRB (Physical Resource Block) utilization for eMBB (enhanced Mobile Broadband) traffic often causes delays to exceed

the 10 ms Packet Delay Budget (PDB). This impacts the overall experience, especially in applications requiring strict real-time feedback like VR and AR environments.

In summary, 5G networks can meet the latency demands of XR use cases under optimal conditions but fall short in high-density environments, where delays can surpass acceptable thresholds and directly impact the user experience.

D. RELIABILITY

In XR applications, maintaining high reliability means ensuring that a significant percentage of video frames (e.g., 99%) are delivered within the target latency to avoid user dissatisfaction. The reliability metric is influenced by several factors, including the density of users and interference from other types of traffic, such as enhanced Mobile Broadband (eMBB) traffic.

In dense urban environments, the 5G network can support approximately 13 XR user devices (UEs) per cell when each user is consuming 30 Mbps of XR traffic. However, as more users are added, the satisfaction rate decreases significantly. For example, with eMBB traffic present, there can be a loss of up to 80% in XR capacity, especially if a strict packet delay budget (PDB) of 10 ms is enforced [36]. By adjusting the PDB to 20 ms, the XR capacity can be maintained, although eMBB throughput must be restricted by 50% to balance the demands of both traffic types [36].

5G URLLC provides the mechanisms to meet reliability requirements, but as user density increases, the performance declines. In high-density scenarios such as urban environments or museums, the motion-to-photon latency in XR applications can reach up to 50–100 ms, which negatively affects the user experience [39]. The ability to sustain high reliability decreases as user congestion increases, primarily due to delays introduced by the growing need to process more user requests.

To mitigate interference, techniques like Inter-cell Interference Coordination (ICIC) and restrictions on the use of eMBB resources have been proposed. These mechanisms aim to protect XR services from degradation due to mixed traffic scenarios, enhancing the performance and reliability of XR under high network load conditions [36].

Summarizing the above, although 5G provides the infrastructure to meet XR's reliability requirements under normal conditions, its performance can significantly degrade in densely populated environments. Techniques such as restricting eMBB traffic or adjusting the packet delay budget (PDB) can help maintain reliability and improve user satisfaction when traffic congestion occurs.

E. XR CAPACITY

XR capacity refers to the maximum number of UEs per cell that can meet the satisfaction threshold under varying network conditions. Satisfaction is typically defined as 99% of the packets being successfully delivered within the set packet delay budget (PDB) requirements. For XR applications, meeting these thresholds is influenced by traffic

types, packet size, and interference from other services like enhanced Mobile Broadband (eMBB). For advanced Metaverse use cases, where the experienced service bit rate is provisioned on at a speed of 225Mbps to allow HD VR streaming [43]. 3GPP has defined the experienced user bit rate for viewport streaming -offloading heavy rendering tasks to a powerful device, which streams only the user's current view to AR glasses via tethering- between 200-2000Mbps [44]. Based on the results reported in Table 3, these bit rates are not yet sustainable by 5G.

In frequency range 1 (FR1), 5G supports between 2 and 5 UEs per cell when each device requires 45 Mbps for XR traffic. This capacity can improve to 3-8 UEs per cell when the traffic demand is reduced to 30 Mbps and staggered traffic scheduling is employed. In frequency range 2 (FR2), the performance is notably better, supporting between 20 and 28 UEs at 45 Mbps, and 26 to 35 UEs at 30 Mbps [39].

However, mixed traffic scenarios, where both XR and eMBB services coexist, can significantly reduce XR capacity. In such scenarios, the introduction of eMBB traffic can cause up to an 80% loss in XR capacity, primarily due to increased inter-cell interference (ICI) from eMBB. Experiments showed that with strict PDBs of 10 ms for XR, this interference poses a major challenge. Capacity can be maintained by employing radio resource management strategies that limit eMBB throughput by around 50%, enabling better XR performance even when both types of traffic coexist. For example, by increasing the XR PDB to 20 ms, the negative impact of eMBB can be mitigated, allowing up to 6 XR devices per cell at 30 Mbps in dense urban environments [36].

Increasing the PDB for cloud gaming services to 15 ms, compared to AR/VR services with 10 ms, also results in a capacity improvement of about 1.3 times. Simulations have demonstrated that by using radio resource management, XR capacity can be maintained, although the presence of eMBB traffic can still reduce the total number of satisfied users per cell to as low as one, particularly in dense urban scenarios [40]. Furthermore, in [45], it is argued that 5G networks, particularly those up to 3GPP Release 17, are capable of supporting many Metaverse use cases. 5G can handle up to 10 simultaneous high-quality real-time XR users, depending on the application and network conditions. However, scaling these services to support more users or highly demanding applications may lead to limitations. To address these issues, 5G-Advanced (5GA), with enhancements introduced in Release 18, incorporates features like flexible Quality of Service (QoS) handling and congestion control mechanisms. These improvements, such as the PDU set optimization and L4S (Low Latency, Low Loss, and Scalable Throughput), are designed to meet the latency, reliability, and throughput needs of more demanding XR and Metaverse applications [45].

In summary, while 5G provides adequate support for isolated or smaller-scale Metaverse applications, its current deployments cannot cope with scaling and underload

conditions. The future enhancements in 5GA (Release 18 and beyond) aim to overcome these limitations, making 5G more suitable for large-scale metaverse services deployment.

F. POWER CONSUMPTION

In the context of power consumption for XR devices, which includes head-mounted displays (HMDs) and AR glasses, several methods have been suggested to meet the stringent requirements for low power consumption (<3W). According to [39], the power consumption for 5G modems used in XR devices is typically around 0.5W, which is within acceptable limits. However, as XR traffic grows and involves low-latency, high-reliability transmissions, reducing power consumption while maintaining service quality becomes challenging.

In addition, optimizing power consumption in XR applications is complex due to the quasi-periodic nature of XR traffic, which requires active management of the modem's sleep states to avoid unnecessary energy use. The authors in [1] discuss several advancements introduced in 5G-A that aim to mitigate power consumption. One such advancement is optimizing Discontinuous Reception (DRX) cycles to better align with the periodicity of media streams, such as a 60 FPS video. This can reduce power consumption without compromising the delivery of traffic. In 5G-A, a notable improvement is the ability to detect the end of a media burst, allowing the network to immediately signal the device to enter a power-saving state. By better aligning DRX cycles with media packet arrival times, especially for media streams with fixed periodicity (e.g., 60 FPS), the system reduces the modem's active time, thus preserving energy.

Even though these optimizations reduce power consumption, challenges arise in dense traffic scenarios. The need to maintain low latency while ensuring devices stay in low-power states under high-load conditions can lead to increased power draw. The authors in [39] highlight that additional innovations in managing power consumption are necessary as the XR user base expands in dense urban environments.

Thus, while the advancements in 5G-A allow for more efficient power usage, particularly with DRX cycle optimization, maintaining this efficiency in high-demand environments remains a significant challenge.

G. POSITION AND TIMING ACCURACY

5G and 5G-Advanced offer significant advancements in tracking precision for AR/VR experiences. According to [41], 5G-A can achieve microsecond-level accuracy (1-10 μ s) by utilizing over-the-air synchronization (OTA), Precision Time Protocol (PTP), and Time Sensitive Networking (TSN). This level of precision supports high-quality, real-time positioning necessary for immersive Metaverse applications and AR/VR environments.

In terms of location accuracy, the authors in [42] propose a method that uses 5G uplink (UL) carrier phase measurements for six degrees-of-freedom (6DoF) tracking.

The results show that this method achieves sub-centimeter positioning accuracy—down to less than 1 cm—and can provide orientation accuracy below 1 degree, even in complex environments such as indoor factories. The authors suggest that this level of precision is achievable at higher frequency bands like 28 GHz, making 5G a robust solution for real-time XR device tracking.

Integrated sensing and communication (ISAC) frameworks are expected to further enhance positioning and awareness capabilities. For example, the authors in [46] demonstrate that mmWave MIMO-OTFS systems can jointly estimate spatial parameters, such as delay, Doppler, angle-of-arrival (AoA), and angle-of-departure (AoD), with high accuracy, even under mobility and noise. While not XR-specific, such mechanisms may serve as foundational components in future 6G networks, enabling real-time scene awareness and spatial context essential for immersive XR and Metaverse applications.

These advancements demonstrate that 5G-A can handle the demanding requirements of XR applications, providing both accurate real-time location tracking and synchronization critical for smooth, immersive experiences.

V. SECURITY AND PRIVACY CONCERNS IN 5G-ENABLED IMMERSIVE APPLICATIONS

A. SECURITY ASPECTS

XR/VR service providers are often not aware of the inherent dangers that come with the usage of Immersive Technologies. The vast number of threats are not limited to conventional cybersecurity threats and vulnerabilities. As immersive technologies and applications tend to adopt a service-based notion, they are vulnerable to typical dangers faced in ICT systems such as malware, phishing and denial-of-service, unauthorized usage, and well-known weaknesses in systems (e.g., unpatched software, weak passwords, and miss-configurations). Depending on the services and the implementation of XR/VR use cases, the attack vector becomes significantly larger and encapsulates threats from verticals. The case, highlighted in this work, is the integration of immersive technologies and advanced networks such as 5G/B5G that follow the MEC paradigm. The authors in [47] discuss advanced networks and XR/VR as separate vectors that each one of them is comprised of different characteristics, threats, vulnerabilities, and countermeasures.

Specifically, 5G/B5G as a standalone enabler of XR/VR use cases comes with severe threats and vulnerabilities, especially due to the trend of network virtualization, shared resources within the MEC concept, and the extended attack vectors that come from the modular components that define such network architectures, as mentioned in Section III. ENISA in [48] identified that vulnerabilities are most common in the categories described below. These categories are filtered out with respect to immersive application applicability in corresponding XR Use Cases.

1) VIRTUALIZATION [49]

Virtualized components must be trusted by the virtualization machine that handles and monitors their proper functionality. The compromising of the VM host is of major importance as it directly affects the availability of residing virtualized functions and services that users or service providers can create (deployed as VMs) to be accessed and used through an XR Use Case. Moreover, the transparency of VMs poses security risks when not handled correctly, leading to reverse process engineering and compromising cryptographic assets.

One effective measure to enhance the security of virtualized components is the use of a Trusted Platform Module (TPM) [49]. The TPM offers secure storage for cryptographic keys and ensures the integrity of virtual machine (VM) components through chains of trust. By encrypting sensitive VM data, TPM adds an additional layer of security, helping to prevent unauthorized access and protect against potential vulnerabilities in the virtualization environment.

2) APPLICATION PROGRAMMABLE INTERFACES (API) SECURITY

Significant emphasis is placed on the development of secure APIs, as they form a critical component of modern systems. The authors in [50] demonstrate this by incorporating robust authentication mechanisms within APIs. It is ensured that access is restricted to authorized users only, thereby preventing unauthorized or potentially malicious entities from exploiting an API. As a result, sensitive data is protected and the integrity of the system is maintained

- **Token-based authentication** - In this approach, unique identification strings (tokens) are assigned to user sessions following successful authentication. These tokens, typically encrypted, are exchanged between the client and server during request-response cycles and remain valid for a predefined time.
- **Cookie-based authentication** - Similar to token-based methods, cookie-based authentication creates a small file (cookie) to store authentication data. This method relies on the framework in which the API is developed, managing authentication on the server side.
- **Key-based authentication** - Unlike token or cookie methods, key-based authentication does not require user credentials for each session. Instead, it uses cryptographic keys (like API keys) to verify user identity, ensuring secure access through cryptographic validation.

Beyond authentication, securing APIs requires additional measures, particularly for Immersive service-based systems like XR environments. Input data, especially complex requests from both data producers and consumers (e.g., for virtual events or remote exhibitions), needs to be sanitized and validated to prevent injection attacks. Many APIs do not mandate cryptographic protocols by default, which leads to the necessity of implementing custom security measures, especially in high-security XR/VR applications. Common

cryptographic solutions include OAuth 2.0, OpenID Connect, and Mutual TLS (mTLS). In mTLS, both parties exchange certificates to authenticate each other and perform a TLS handshake using asymmetric cryptography, ensuring secure, encrypted communication.

3) 5G PRIVATE NETWORK SECURITY

Since the usage of immersive applications for cultural heritage is a privilege for private 5G networks, this category falls under the latter. We consider that these 5G network attacks do not fall under a broad network that integrates other networks of previous generations and their concurrent usage. The paper in [51] distinguishes attacks based on the protocols that are used.

In particular, the authors mention that attacks in the RRC protocol can lead to subscriber ID tampering, user authentication bypass, and DoS/DDoS attacks on the base station. In an immersive environment, where users are continuously interacting with the virtual space, adversaries could exploit RRC vulnerabilities, resulting in users being disconnected or experiencing latency issues that render the immersive experience useless [51]. By tampering with the RRC signaling messages, attackers can impersonate users, leading to unauthorized access to exclusive content or personal user data leakage. Similarly, by flooding a private 5G network with fake RRC connection requests, attackers could overwhelm 5G base stations and prevent legitimate users from establishing or maintaining their connection. This could cause users to experience severe interruptions in their experience, disconnecting them from the virtual environment.

As users move through different virtual environments or interact with augmented reality content, NAS protocol vulnerabilities could be exploited to track their location and movements [52]. NAS-based attacks could also lead to session hijacking, allowing attackers to take control of the user's session, modify their interactions with the heritage site, or cause authentication bypasses.

Moreover, the authors in [53] explain that GTP protocols managing data tunneling could be compromised, allowing attackers to intercept or manipulate the real-time data that generates the immersive environment. For example, in a VR heritage tour, this could result in corrupted or incomplete virtual environments. Man-in-the-middle (MitM) attacks can potentially alter the historical content presented to users, severely undermining the educational and historical accuracy of the application.

Significant importance also falls on the diameter protocol which is an IP-based protocol for authentication, authorization, and accounting, providing high levels of QoS to the 5G network infrastructure. Similarly, since the Diameter protocol handles critical services like user authentication, an attacker could hijack user sessions, potentially exposing sensitive information or causing financial damage if the immersive service includes premium features. Replay attacks may also disrupt real-time interactions within the heritage experience, degrading the quality of the engagement [54].

The work in [53] showcases that such protocol-based attacks could affect most of the 5G NSA network domains, varying from UEs, RAN, Core Network, and even external Networks connected to a private 5G infrastructure. Precisely, the authors explain that especially GTP-based and Diameter protocol attacks can directly target Core Network functions such as the PCF, AMF, and UPF.

B. PRIVACY ASPECTS

XR Devices have a great number of smart sensor capabilities, capturing user, environment, and bystander information from actions, physiology, visual, audio, and haptic input from the sources. These sensors are fundamental for the proper functionality of XR technologies, but at the same time capture a considerable amount of data. Data available to XR devices amplify insight generation capabilities and, in many cases, sensitive information can be either estimated or inferred by reverse engineering either personal or environmental attributes such as user sexuality, gender, location, age, height, and even personal preferences [55]. In our analysis, we identify the following privacy threats related to Smart Culture Applications:

- **Anonymity and Privacy in XR.** The issue of user privacy and network security is magnified with the use of XR technologies, especially in the context of museums and interactive exhibits. The incorporation of personal data into these applications may lead to potential disclosure to third parties, raising significant privacy and security concerns. It is crucial, therefore, to extensively investigate these issues and develop robust personal data protection protocols. This could involve implementing advanced encryption techniques, anonymization of data, stringent access control mechanisms, and educating users about potential risks and their rights [56], [57]. It's essential also to ensure compliance with data protection laws such as the General Data Protection Regulation (GDPR).
- **Pervasive Data Collection.** XR applications are often multi-user and demand remote collaboration. The immersive interaction between avatars requires vast amounts of information derived from user profiling activities [58]. The authors explain how unreasonably granular activities such as the forming of facial expressions, hand movement, speech, biometric features, and even brain-wave patterns are commonly tracked between advanced XR technologies and can lead to person identification. Motion sensors built within cameras in AR helmets such as Oculus, are able to track head directions, position, and environment in real time with considerable accuracy levels. Only considering the level of personal information these data possess, form great risks for businesses that collect them and especially for data subject individuals. In addition, the authors in [59] discuss that captured data from XR and enabling devices can form detailed invasive privacy models by combining pervasive data collection and

interactive behavioral metrics, such as 'walk-through and click-through rates'. The aforementioned models are also able to derive a subject's subconscious behavior, deliberate intent, learning and memory, social cues, and sensory profiles without consent.

- **Leakage in Data Transfer.** In many cases, wearable equipment, used for XR use cases and applications, transfers personally identifiable information via wired and wireless communications, especially within 5G ecosystems. As a result, it is necessary to ensure secure data transfer for sensitive data. Although communication is encrypted and information is transmitted with confidentiality, adversaries are still able to access raw data by eavesdropping (MitM attack) on the specific channel and still track the user's location via differential and advanced inference attacks [60].
- **Data Leakage during Storage.** Storing confidential and sensitive data, such as user profiles, for a large number of users in cloud servers or edge devices can give rise to concerns regarding privacy breaches. Furthermore, the work in [61] highlights that shifts to user-centric data storage models allow the usage of adversary analytical tools that can steal sensitive information. Moreover, as B5G adopts cloud-native concepts (SBA), various data-at-rest attacks are also affecting XR applications [62]. These include linkage attacks where unauthorized access could grow with linked XR data sources, homogeneity attacks, targeting k-anonymity techniques, and traditional data manipulation attacks in which modified data could be altered by service-specific exploits. One notable concern is the potential for privacy disclosure through frequent queries, as malicious actors might employ differential attacks to deduce users' private information. Additionally, the security of cloud/edge storage may be compromised through distributed network attacks, allowing unauthorized access to the stored data. A similar incident occurred in [63] when a customer database of Second Life, a Metaverse game, fell victim to hacking, resulting in the exposure of sensitive user data. The compromised information included unencrypted usernames and addresses, along with encrypted payment details and passwords. This incident serves as a poignant example of the real-world consequences of insufficient privacy measures in cloud/edge storage systems.

VI. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

A. XR CAPACITY LIMITATIONS

Capacity remains a significant challenge in supporting high bit rates for XR applications, especially with increasing numbers of users. Radio resource management and interference mitigation techniques are key to addressing capacity issues. The PDU-set scheduling algorithm proposed by [36] is one such approach, focusing on efficient resource allocation between XR and eMBB services. Similarly, [64] introduced

a time and frequency domain scheduling technique for URLLC and eMBB coexistence. This solution improved eMBB throughput by 90% while reducing URLLC latency by 29% [64].

Another method proposed [65], utilizes Radio Environment Maps (REMs) to coordinate interference, improving system throughput by 72% in ultra-dense environments [65]. Furthermore, the authors in [66] presented an interference vector judgment-based user clustering method to enhance capacity, particularly for cell-edge users in ultra-dense networks. Another challenging field is mMIMO. In [67], the authors investigated the coexistence of eMBB and URLLC traffic in multi-cell massive MIMO systems. Their work compared puncturing (PUNC) and superposition coding (SPC) strategies, showing that SPC offers higher spectral efficiency while still meeting URLLC reliability needs.

Future studies are expected to focus on further reducing inter-cell interference and improving scheduling algorithms as 6G systems aim to manage much higher user densities and more complex service requirements. With the increasing demand for mixed services like XR and eMBB, future network releases must refine these methods to sustain high-capacity performance in ultra-dense and dynamic environments.

B. POWER CONSUMPTION

Reducing power consumption in XR devices, such as AR headsets and VR goggles, remains a significant challenge. Currently, most smartphones cannot support full metaverse platforms by themselves [43]. Recent studies highlight various methods aimed at optimizing energy efficiency while maintaining the performance needed for immersive applications. Current trends focus on using embedded AI and task-offloading techniques to minimize the energy demands of real-time processing. For example, the paper in [68] discusses the use of embedded AI within 6G networks to optimize power consumption. By allowing devices to offload computationally intensive tasks to the network edge or cloud, this approach reduces energy use without sacrificing performance. Similarly, the authors in [69] explore optimizing power consumption through energy-aware MEC. Their work highlights dynamic offloading and adaptive task scheduling as methods to manage power demands more effectively in next-generation networks, emphasizing energy-efficient offloading for resource-intensive XR applications.

Future research should continue exploring these methods, with a particular focus on balancing energy efficiency and performance to meet the growing demands of next-generation XR applications. This includes improving AI offloading mechanisms, better managing sleep-wake cycles of devices, and enhancing edge computing strategies to allow XR devices to operate efficiently over extended periods.

C. XR FOR DIGITAL TWINS

Digital Twin Networks (DTNs) offer a real-time virtual replica of physical networks, facilitating the design,

simulation, and management of complex 5G and 6G wireless environments. These systems enable continuous feedback loops between the digital and physical network layers, allowing for real-time optimizations and decision-making. In 6G, DTNs are anticipated to support various use cases such as network simulation, predictive maintenance, and AI-driven operations, making network operations more efficient and adaptive [70]. By providing accurate simulations and predictive insights, DTNs help network operators optimize resource allocation and plan deployments more effectively, improving overall network performance and reliability in dynamic conditions. In addition, DTNs enable operators to explore “what-if” scenarios, ensuring network robustness against potential failures or misconfigurations, ultimately supporting more resilient and efficient 6G networks [70].

The future of Digital Twins (DTs) research focuses on several important areas aimed at optimizing their integration into complex systems. A key challenge is the issue of ownership, where the physical object and its digital representation might be controlled by different entities, raising questions around data privacy, control, and storage, especially in multi-component systems like industrial setups [71]. Ensuring real-time synchronization between the physical object and its DT requires ultra-low-latency, highly reliable communication, which current technologies cannot provide, making further advancements necessary. The creation of DTs for entire 5G networks will expand network exposure (Fig. 3) and may involve the extension of the 5G Core to support DTs, enabling the development of comprehensive digital replicas of these complex systems.

D. SECURITY AND PRIVACY IN THE METAVERSE

As mentioned in Section V, the introduction of 5G has expanded the attack surface due to the integration of various technologies. Ongoing research is exploring new solutions to address the resulting security threats. Edge processing—used to offload tasks for advanced XR use cases—highlights the need for trusted execution environments [72]. Furthermore, the authentication of users is a key area of focus, particularly in immersive and collaborative Metaverse scenarios, which raise serious privacy concerns. Another concern is the monolithic identity management model used by centralized authorities, which is prone to single points of failure, has limited scalability, and restricts user control, [73]. This highlights the need for new decentralized identity models that offer greater resilience and empower users with more control over their personal data and authentication processes.

A promising approach to tackle these concerns is the implementation of self-sovereign identities (SSI) and credential wallets. SSI allows individuals to retain control over their personal data, sharing only the necessary details with service providers while keeping sensitive information private. Credential wallets, like the EU’s Digital Identity Wallet Initiative, securely store personal documents and identity credentials, enabling users to authenticate themselves without disclosing excessive information.

In XR environments, where extensive data collection occurs—from biometric data to behavioral patterns—SSI empowers users to manage who can access their information and how it’s utilized. These decentralized identity systems reduce dependence on centralized authorities, lowering the risks of data breaches and unauthorized tracking. Integrating SSI and credential wallets into XR use cases enhances user privacy, ensures compliance with regulations like GDPR, and strengthens trust in immersive applications by giving individuals more control over their digital identity.

Additionally, 3GPP has considered credential wallets as part of its feasibility study on localized mobile Metaverse services [44]. The EU’s Digital Identity Wallet aims to provide a unified, secure platform for storing personal data and documents, such as IDs and certificates, while offering users the ability to control data sharing across services in the EU. Though primarily designed for Europe, this initiative’s principles could be extended globally, helping to establish trust in digital services and facilitating secure e-commerce and e-government interactions.

VII. CONCLUSION

This paper examined the integration of XR applications within 5G networks, focusing on how the combination of advanced network infrastructure and immersive media can reshape various sectors. We provided an overview of 5G’s service-based architecture, highlighting how MEC improves XR support by reducing latency and bringing processing power closer to users. The role of 5G network analytics was discussed in relation to XR use cases, offering a path toward more efficient network management for these demanding applications.

Through exploring XR’s application in cultural heritage and education, the paper demonstrated the potential of this technology to enhance user experiences and access to digital content. The analysis also addressed security, privacy, and ethics concerns surrounding XR environments, identifying specific risks and the need for stronger data protection measures as these technologies become more widely adopted.

While 5G presents a strong foundation for XR, challenges remain in areas like scalability under heavy network usage, power consumption, and the complexity of handling user data. As 3GPP begins its study items on 6G, it remains to be seen whether the fully immersive Metaverse experience will be realized in future 6G networks.

In terms of future research, several directions emerge from this work. There is an imperative need for further investigation into the integration of advanced functionalities in 6G, including AI-driven predictive network analytics and enhanced security mechanisms tailored specifically for XR scenarios in smart culture applications. In addition, addressing scalability issues through efficient resource allocation and optimized MEC deployments could significantly improve user experience in high-density environments. Finally, exploring novel identity management frameworks, such as decentralized Self-Sovereign Identities and robust

privacy-preserving mechanisms can ensure secure and personalized user interactions within the emerging Metaverse ecosystem.

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